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ORIGINAL ARTICLE NON-SURGICAL AESTHETIC

Advanced Techniques for the Use of Neurotoxins in Non-Surgical Facial Rejuvenation

Jason E. Cohn[1] · **Timothy M. Greco**[2]



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Abstract
Background Neurotoxins initially were used to treat hyperfunctional rhytids of the face, but now have been expanded to improve facial shaping, correct facial asymmetry and even improve skin texture and tone.
Methods The clinical approach to non-surgical facial rejuvenation is approached into four anatomical regions: the upper face, midface, lower face and neck.
Results The key muscles of the upper face include frontalis, orbicularis oculi, corrugator supercilii, procerus, depressor supercilii and temporalis. The muscles in the midface to be discussed include the levator labii superioris, levator labii superioris alaeque nasi, depressor anguli oris, depressor septi nasi and nasalis. Treatment of the lower face focuses on the orbicularis oris, mentalis, depressor anguli oris and masseter muscles. Finally, treatment of the neck region will be reviewed with emphasis on platysmal bands and necklace lines as well as the Nefertiti lift.
Conclusions Non-surgical facial rejuvenation using neurotoxins should be performed safely and effectively in order to avoid and treat complications.
Level of Evidence V This journal requires that authors assign a level of evidence to each article. For a full description of these Evidence-Based Medicine ratings, please refer to the Table of Contents or the online Instructions to Authors www.springer.com/00266.

Keywords Facial rejuvenation · Injectables · Facial plastic surgery · Cosmetic surgery · Neurotoxins · Botulinum toxins

Introduction

The two most popular minimally invasive procedures for facial rejuvenation include botulinum toxins and hyaluronic acid [1, 2]. The first report of botulinum toxin use for cosmetic purposes was published in 1990 by Carruthers and Carruthers [3]. Neurotoxins initially were used to treat hyperfunctional rhytids of the face, but now have been expanded to improve facial shaping, correct facial asymmetry and even improve skin texture and tone [2, 4]. The pathogenesis of the wrinkle is important in determining how effective Botox will be in effacing it. When the rhytid is primarily or at least significantly caused by muscular action deforming the overlying skin, Botox can be an extremely effective treatment in the lower face [5]. More specifically, neurotoxins can help with brow shaping/positioning, eye shaping, nasal tip lift, jawline shaping, chin shape and definition, masseter reduction, eversion of the lips and platysmal bands [6]. The cosmetic use of botulinum toxin is not only limited to the upper face. Its use in the midface, lower face and neck has become increasingly popular because physicians feel more confident in its efficacy and safety profile [7–8]. Patients have also demonstrated increased satisfaction rates and quality of life after receiving botulinum toxin treatment [9–10].

✉ Jason E. Cohn
jasoncoh@pcom.edu

Timothy M. Greco
timg@drgrecoface.com

[1] Department of Otolaryngology-Head and Neck Surgery, Division of Facial Plastic Reconstructive Surgery, Philadelphia College of Osteopathic Medicine, 4190 City Line Avenue, Philadelphia, PA 19131, USA

[2] Center of Excellence in Facial Cosmetic Surgery, 2 Bala Plaza, PL-15, Bala Cynwyd, PA 19004, USA

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It is important that when injecting the facial musculature that one has a firm grasp of the differences between the FDA-approved neurotoxins, which include Botox™ (onabotulinumA), Xeomin™ (incobotulinumA) and Dysport™ (abobotulinumA) [4]. In addition, there is a new formulation Jeuveau™ (prabotulinumtoxinA) which is indicated for the temporary improvement in the appearance of moderate to severe glabellar lines [11]. However, its use has expanded to other facial regions. One recent study demonstrated a trend toward treating the superolateral orbicularis oculi and the depressor anguli oris muscles [9].

It is not only important to understand the anatomy of the muscles but also the product of choice, the dilution of the product, the characteristics of needle used for placement (length and gauge), and the angle, extrusion force and depth of injection. All of these factors are critical to the final result and help to avoid complications. A rough estimate of 1 unit onabotulinum, incobotulinum or prabotulinum corresponds to 2.5–5 units of abobotulinum [6–7, 12–16]. Generally speaking, the effect of botulinum toxin has been noted to begin 24–48 h after injection. Furthermore, the average duration of action is approximately 2 months, with a range of 0.5–4 months [14, 16]. The shortest response expectedly occurs with lower dosages [16], while repeated injections show a longer duration of effect [14]. Larger muscles often require larger doses of neurotoxin, and the volume of the dose should be adjusted to the desired diffusion of toxin. More concentrated doses diffuse less and should be used to target smaller muscles [14].

The use of botulinum toxin for cosmetic purposes can be met with complications [17–19]. It is advisable to use the smallest effective dose in the upper third of the face to try to avoid unwanted outcomes such as asymmetry, brow or lid ptosis and mouth asymmetry. An important consideration when choosing which neurotoxin to use in each anatomic location is the spread of effect. Onabotulinum and incobotulinum have comparable spread, while abobotulinum has a greater spread of effect. Unwanted spread into untargeted muscle groups can lead to undesirable clinical effects. A neurotoxin with a low and predictable spread would be preferable for treating the glabellar muscle complex, and a larger spread of effect could optimize treatments for the forehead lines and crow's feet. Particularly in the lower one-third of the face, injection sites are very close to untargeted muscles; therefore, it is important that the neurotoxin does not spread [12, 13]. Others prefer higher reconstitution volumes and injection volumes per unit dose when they desire greater toxin spread [20].

Some adverse effects that should be discussed with the patient prior to injections should include the possibility of unexpected loss of strength or muscle weakness, injection site reactions, bleeding/bruising, headaches, trouble swallowing or breathing, double or blurred vision, ptosis of eyelids and/or brows, dysphonia, dysarthria and loss of bladder control. Abobotulinum should be avoided in patients with a cow's milk allergy. Additionally, individuals with peripheral motor neuropathic diseases such as amyotrophic lateral sclerosis or neuromuscular junctional disorders (e.g., myasthenia gravis or Lambert–Eaton syndrome) should be monitored particularly closely when given botulinum toxin because they may be at increased risk of clinically significant effects including severe dysphagia and respiratory compromise [14, 17, 20–21].

There are certain measures to be taken in order to avoid complications and adverse events. Generally speaking, patients are advised to avoid anticoagulants, platelet inhibitors (i.e., aspirin), NSAIDs and/or other blood-thinning compounds unless medically necessary for 2 weeks, as well as vitamin E, herbal supplements and fish oil in order to minimize bruising. While injecting, practitioners should wear magnification and use bright illumination to avoid blood vessels. They should also avoid injecting below the periosteum to prevent headaches. Post-procedure, patients should avoid palpating treatment areas for several hours. Alcohol intake and exercise should be avoided for 2 h to prevent excessive perfusion of injected muscle which could washout the product. Patients should also exercise muscle groups injected with botulinum toxin every 15 min for 2 h post-injection [6, 20, 22].

Materials and Methods

The clinical approach to non-surgical facial rejuvenation is approached into four anatomical regions: the upper face, midface, lower face and neck. The patients featured in this manuscript underwent informed consent prior to treatment and provided permission to publish their images to the authors. The patient images were conducted in a private practice setting, with the intent to maintain consistency and standardization. Unfortunately, this was not always possible due to patients' preferences for makeup, hairstyles and accessories. This work does not contain any studies with human participants or animals performed by any of the authors.

Results

Specific anatomical considerations, measures, dosages and complications as they relate to particular muscle groups will be discussed throughout this paper. Injection charts have been illustrated to assist providers (Tables 1, 2, 3 and 4).

Table 1 Neuromodulators in the upper face

Muscle	Onabotulinum	Incobotulinum	Abobotulinum
Orbicularis oculi (lateral crow's feet)	6–12U	6–12U	18–36U
Orbicularis oculi (jelly roll)	1–2U	1–2U	3–6U
Glabellar complex (women)	15–25U	15–25U	45–75U
Glabellar complex (men)	25–40U	25–40U	75–125U
Frontalis (women)	5–12U	5–12U	15–36U
Frontalis (men)	12–20U	12–20U	36–60U

U units

Table 2 Neuromodulators in the midface

Muscle	Onabotulinum	Incobotulinum	Abobotulinum
Zygomaticus major	2U	2U	3–6U
Levator labii superioris	1–4U	1–4U	3–12U
Levator labii superioris alaeque nasi	1–2U	1–2U	3–6U
Depressor septi nasi (midline)	3–6U	3–6U	9–18U
Dilators of nares	2–6U	2–6U	6–18U

U units

Table 3 Neuromodulators in the lower face

Muscle	Onabotulinum	Incobotulinum	Abobotulinum
Orbicularis oris (upper)	3–4U	3–4U	9–12U
Orbicularis oris (lower)	2U	2U	6U
Depressor anguli oris	3–5U	3–5U	9–15U
Depressor labii inferioris	3–4U	3–4U	9–12U
Mentalis	4–10U	4–10U	12–30U
Masseter (each side)	10–40U	10–40U	30–120U

U units

Table 4 Neuromodulators in the neck

Muscle	Onabotulinum	Incobotulinum	Abobotulinum
Platysma[a]	2.5U	2.5U	7.5U
Necklace lines[b]	1U	1U	3U

U units

[a]Every 1.5 cm along vertical bands (do not exceed 50U/50U/150U)

[b]Every 1 cm along flexion crease

Discussion

Upper Face

Neurotoxins are commonly used in the upper face to minimize wrinkles due to muscular contraction. The orbicularis oculi is a very large muscle that extends far out beyond the orbital rim and creates lateral canthal lines known as crow's feet. The lateral injections involved with treatment of the crow's feet should not extend more medial than the extent of the orbital rim. Injections placed medial to this critical anatomic landmark can result in diplopia and/or strabismus from paralysis of the lateral rectus muscle. Treatment of the crow's feet typically requires up to 15 units of onabotulinum, incobotulinum, prabotulinumtoxin, or up to 45 units of abobotulinum (per side) are injected evenly in 3–4 sites within the muscle at least 1 cm lateral to the lateral orbital rim to minimize intraorbital complications (Fig. 1a, b) [6, 17, 23–24]. In one

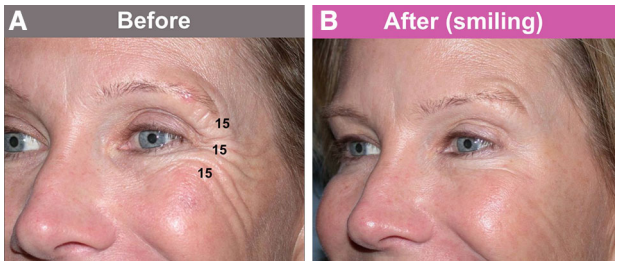


Fig. 1 Before (**a**) and after (**b**) appearance after treatment of crow's feet with neurotoxin

study, abobotulinum had a trend toward greater improvement than onabotulinum at day 2 and a statistically significant greater improvement at days 4 and 6 when looking at maximal contraction of lateral orbital rhytids [25].

When it comes to the aging face, deeper lines or furrows can be classified as either dynamic or static. Dynamic lines appear with movement, while static lines are unchanged with muscle movement; however, dynamic lines eventually become static. Dynamic lines are a result of hyperfunctional muscles, while static lines represent hyperfunctionality as well as skin damage from aging [6]. Botulinum toxin and filler are commonly co-administered to treat dynamic and static lines, respectively [26]. When trying to treat lines in the lower crow's feet region that occur after the injection of neurotoxin, it is important that the patient understand that an interaction between adynamic and dynamic areas of the face has been created. So the junction between the dynamic and adynamic areas creates single or multiple lines that the patient will express concern after they have been treated with a neurotoxin [27]. It should be explained to the patient that these lines can be treated with a flexible low G' hyaluronic acid injection to create increased volume to the skin which then decreases the static and dynamic rhytids in this particular situation [28]. This is best demonstrated in the before and after photographs of the patient in Fig. 2a, b.

The orbital orbicularis oculi is responsible for the forceful closure of the palpebral fissure which is a conscious protective mechanism for the eye. In contrast, the pretarsal and preseptal portions are responsible for the natural blink reflex. The pretarsal portion is in intimate contact with the lower eyelid skin. It is the contraction of this muscle with smiling that creates several unappealing aesthetic facial features [29]. The relative closure of the palpebral fissure can be seen in the before picture of this patient (Fig. 3a). This patient has a relative hypertonicity of her pretarsal orbicularis oculi of both the left upper and lower eyelids. The contraction of this muscle is creating ptosis to the upper eyelid and elevation of the lower lid. Therefore, the pre-injection MRD-1 (the distance from the light reflex of the pupil and the upper ciliary margin) and MRD-2 (the distance of the light reflex of the pupil and the lower ciliary margin) are decreased (Fig. 3b). In this particular patient, a favorable result was achieved by injecting 0.5 units of onabotulinum to the upper eyelid pretarsal orbicularis oculi muscle medially and laterally and 2 units mid-pupillary in the lower pretarsal muscle (Fig. 3c). Theses injections involve subcutaneous placement with the tip of the needle just encroaching the surface of the skin of the upper eyelid. Placing lateral tension on the upper and lower eyelid skin and applying gentle traction with the non-injecting hand with the patient looking downward and upward, respectively, allow for a safe and accurate injection [6].

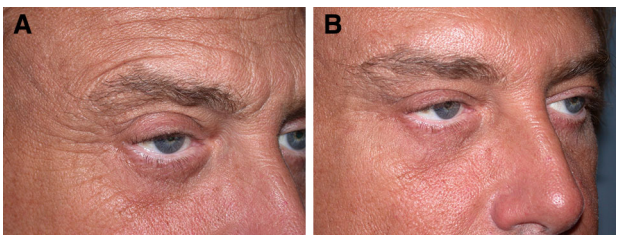


Fig. 2 Before (**a**) and after (**b**) pictures of a patient treated concomitantly with neurotoxin and hyaluronic acid to treat dynamic and static lines, respectively

Another major indication for neurotoxin in the upper face is brow rejuvenation. It is important to note aesthetic and anatomic differences between men and women when planning facial rejuvenation. The ideal male brow has very little variation in thickness or shape. It is placed at the supraorbital rim and is fairly flat. In contrast, the ideal female brow is located at or slightly above the supraorbital rim at the glabella. The brow arch is located above the rim to a point found between the lateral limbus and lateral canthus with a gentle downslope of the lateral one-third. The beginning and ending points of the female brow should be even horizontally [6, 17, 30]. Brow descent can be prevented through a detailed examination, noting the pre-procedural brow position. A balance must then be achieved between the post-injection muscular forces of the frontalis, orbicularis oculi and glabellar complex to maintain an aesthetically pleasing brow position and shape. The muscles of the upper third of the face can be divided into eyebrow elevators and depressors. The frontalis is the sole muscle responsible for elevation of the eyebrows. The corrugator supercilii, depressor supercilii, procerus and the orbital portion of the orbicularis oculi are primarily responsible for depression of the eyebrow [3, 6, 24]. These anatomical relationships have been confirmed with the study of cadavers [31].

The recommendation for the treatment of frontalis is four to nine injection points spaced at least 1 cm apart [6]. Injections should be performed in a "Jacob's ladder," or staggered distribution, at least 1.5–2.0 cm above the brow margin to prevent brow ptosis and asymmetry [2]. The muscle typically requires 5–20 units of onabotulinum, incobotulinum, prabotulinum or 15–60 units of abobotulinum [6, 24]. If the lateral fibers of the frontalis muscle are not injected, patients may develop "eyebrow commas" which is referred in the literature as Mr. Spock, Mephisto or Jack Nicholson appearance [23, 32] (Fig. 4). If the lateral aspect is injected too low or with too many units, it may cause the brow to droop or sag, giving a tired or heavy look. Injection technique for the brow may vary by injector, with some placing 4–6 depots of neurotoxin in the

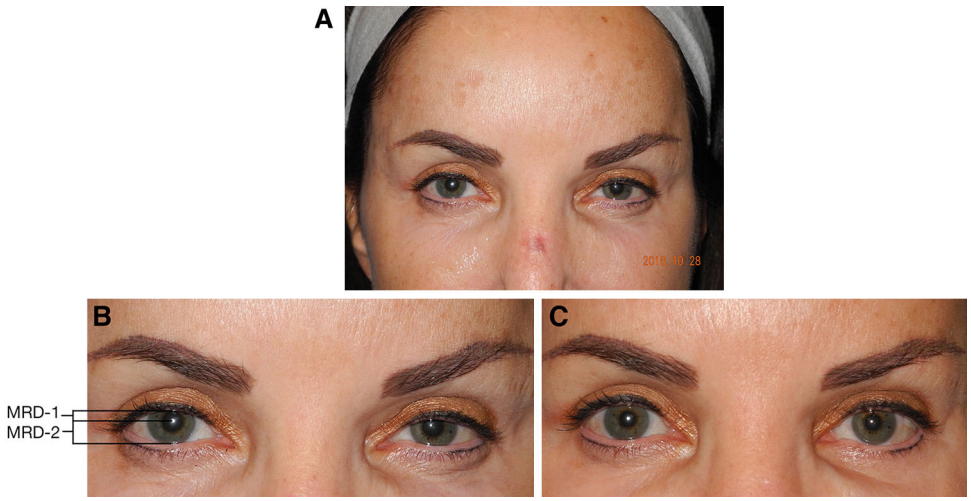


Fig. 3 (**a**) Unappealing ptosis to the upper eyelid and elevation of the lower eyelid due to relative closure of the palpebral fissure due to contraction of the orbital orbicularis oculi muscle with smiling; (**b**) contraction of the pretarsal orbicularis oculi creating ptosis to the upper eyelid and elevation of the lower lid resulting in a decreased MRD-1 and MRD-2; (**c**) favorable improvement in MRD-1 and MRD-2 by applying neurotoxin to the upper eyelid pretarsal orbicularis oculi muscle medially and in the lower pretarsal muscle laterally



Fig. 4 Failure to inject the lateral fibers of the frontalis muscle resulting in "eyebrow commas" which produces a Mephisto or Jack Nicholson appearance

main body of the frontalis. More neurotoxin should be placed centrally in the frontalis while less laterally to achieve recruitment and resultant lateral brow height [6, 24].

Lateral brow descent is also partially caused by the downward pull of the orbicularis oculi muscle. Paresis of the lateral portion of the orbicularis oculi allows for an elevation of the tail of the brow. Brow descent can be prevented through a detailed examination, noting the pre-procedural brow position [6, 24]. A simple technique for lateral brow elevation involves an injection of 1–3 units of onabotulinum, incobotulinum, prabotulinum or 3–9 units of abobotulinum at a single point into the lateral brow depressor (superolateral orbital orbicularis oculi) which weakens it [6]. It is advisable to palpate the superolateral portion of the orbicularis oculi in animation (smiling) to identify the point at which neurotoxin should be injected because eyebrow position varies considerably, especially in women who tweeze their brows as well as patients who have tattooed brows [33]. The treatment of brow ptosis with neurotoxin (before and after) can be best illustrated in Fig. 5.

The glabellar complex comprises the medial brow depressors: procerus, corrugator supercilii, depressor supercilii and the orbicularis oculi muscles. Injection of the procerus should include 1 injection site above and below

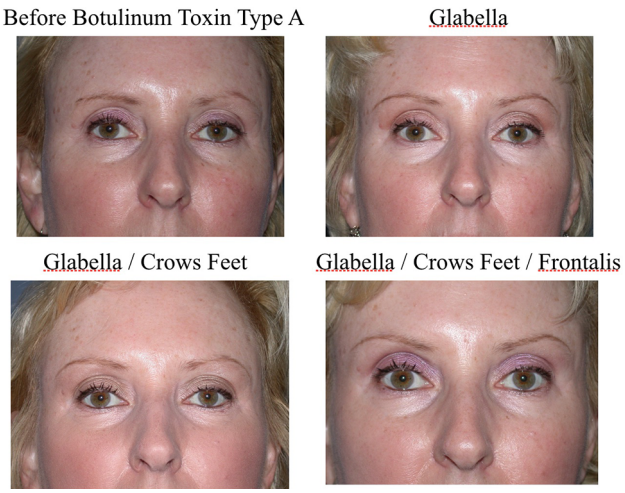


Fig. 5 Neurotoxin injection to brow depressors resulting in brow elevation

the horizontal rhytids in the region of the radix. The injection should be superficial since the muscle runs immediately subcutaneously and is usually 3–6 units of onabotulinum, incobotulinum, prabotulinum or 9–18 units of abobotulinum [6, 24].

The injection of the corrugator supercilii is more complex. Treating this muscle close to the origin requires a deeper injection to capture its medial portion and the depressor supercilii muscle. Laterally, the injection of the corrugator supercilii should be slightly more superficial as the muscle approaches the dermis. The corrugator supercilii typically requires 16–20 units of onabotulinum, incobotulinum, prabotulinum or 48–60 units of abobotulinum. The glabellar complex typical requires 20–25 units of onabotulinum, incobotulinum, prabotulinum or 60–75 units of abobotulinum with 1 injection site into the body of the procerus and 1–2 injection sites (on each side) for the corrugators. The frontalis at its inferior border interdigitates with the brow depressors. As a result of this interdigitation, there can be some spread of neurotoxin to the frontalis when the depressor complex is injected. For this reason, it is prudent to use a concentrated neurotoxin (1–2 cc reconstitution) and to inject low on the procerus and medial belly of the corrugators. The lateral aspect of the corrugator supercilii should be injected just medial to its dermal insertion point and the mid-pupillary line. This technique prevents spread to the medial inferior frontalis, thus preventing medial brow ptosis [2, 6, 24]. Neurotoxin injection to the medial brow depressors is best illustrated in Fig. 6a, b.

During smiling and animation, contraction of the pretarsal orbicularis oculi diminishes the palpebral aperture. This can produce a "jelly-roll" deformity where the contracted muscle produces a subciliary bulge [34]. It can be treated by injecting 2 units of incobotulinum, onabotulinum, prabotulinum or 6 units of abobotulinum, 3 mm below the mid-pupillary line immediately subcutaneous [6]. Treatment results are best illustrated in Fig. 7a, b. This treatment can assist in reducing the muscle bulge as well as open up the palpebral aperture (distance between the upper and lower lids) at rest and when smiling, producing a more

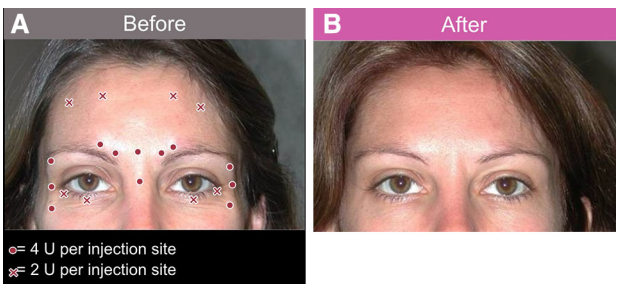


Fig. 6 Neurotoxin injection technique to the medial brow depressors

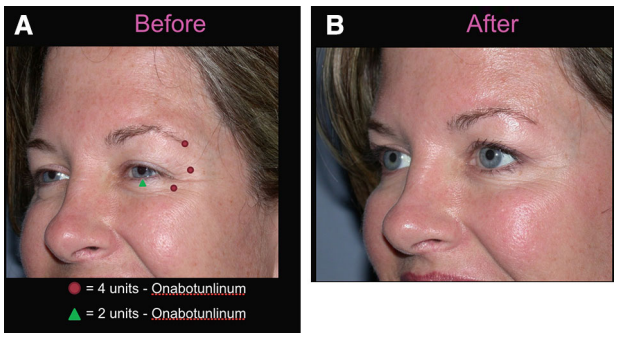


Fig. 7 Before (**a**) and after (**b**) appearance after treatment of the "jelly roll" with neurotoxin

wide-eyed, alert appearance. When evaluating for this kind of injection, a simple physical examination is helpful to determine if the patient is a candidate for such an injection. A snap test is merely a downward traction on the lower eyelid with the index finger and observing how quickly the lower eyelid returns to its normal position [34]. All three neurotoxins can be used on the jelly roll; however, onabotulinum and incobotulinum are better alternatives to be used when correcting upper eyelid ptosis. In contrast, abobotulinum may lead to complications in this area due to a greater spread of effect. This treatment can assist in reducing the muscle bulge as well as open up the palpebral aperture (distance between the upper and lower lids) at rest and when smiling, producing a more wide-eyed, alert appearance [6].

One major side effect of using neurotoxins in the upper third of the face is eyelid ptosis. The incidence of ptosis after such injections initially was reported to be approximately 5%, but recent studies have shown a reduced incidence of this complication [35–36]. Ptosis can occur up to 2 weeks after injections. Ptosis results from migration of the neurotoxin to the levator palpebrae superioris muscle. The levator is the only muscle responsible for opening the eyelid. To avoid ptosis, injections should occur at least 1 cm above the superior orbital rim and should not cross the mid-pupillary line. Apraclonidine 0.5% (Iopidine™) eyes drops are recommended for treating neurotoxin-induced ptosis. Apraclonidine is an α2-adrenergic agonist, which causes Müller muscles to contract, quickly elevating the upper eyelid 1–3 mm [37]. An α2-adrenergic agonist, naphazoline, can also be for this indication. Resolution of neurotoxin-induced eyelid ptosis can be seen in the before (Fig. 8a) and after (Fig. 8b) photographs of this patient after application of 2 drops of naphazoline to the left eye.

A form of reshaping of the forehead with neurotoxins can be accomplished with treatment of temporalis muscle hypertrophy [38]. Although this muscle is not one of facial expressions and is a muscle of mastication, hypertrophy can create an unusual shape to the forehead [39] as seen in

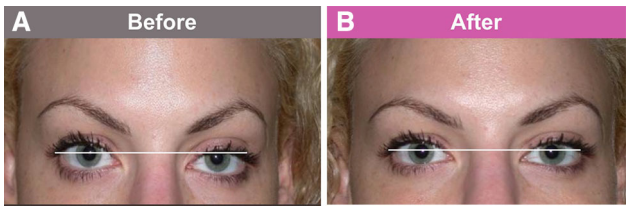


Fig. 8 Resolution of neurotoxin-induced eyelid ptosis can be seen in the before (**a**) and after (**b**) photographs of this patient after application of 2 drops of naphazoline to the left eye

the patient referenced below. This patient benefited from injection of anterior hypertrophy of the temporalis to create a smooth, fresher and rested appearance with a better shape to her brow complex. This can be accomplished by injecting 4 units of onabotulinum, incobotulinum, prabotulinum or 12 units of abobotulinum into three injection points on each side [6] (Fig. 9a–e).

Midface

The midface, which is essentially defined as the region from the zygomatic arch and malar eminence to the nasolabial fold, is primarily an aesthetic region of the face enhanced by fillers. However, neurotoxins have begun to play an increasingly important role in the aesthetic



Fig. 9 A patient being treated for temporalis muscle hypertrophy—before resting (**a**), before clenching (**b**), injection technique (**c**), after resting (**d**), after clenching (**e**)

enhancement in this region. One muscle in the midface is the nasalis which is responsible for centrally and vertically oriented dynamic rhytids known as "bunny lines" [6, 40]. This muscle can be treated with 2–3 units of onabotulinum, incobotulinum, prabotulinum or 6–9 units of abobotulinum on each side of the nose, midway between cheek–nose junction and nasal dorsum [6] (Fig. 10a). Injections that are performed too low can result in lip ptosis and smile asymmetry due to paresis of the levator labii superioris alaeque nasi and levator anguli oris muscles. Caution should also be taken not to over inject superiorly, which can result in medial rectus paresis and epiphora [2]. The results of this injection can be best displayed in Fig. 10b, c.

Another muscle to be discussed in this region is the depressor septi nasi. This muscle creates ptosis of the nasal tip with smiling (animation ptosis), as well as shortening of the upper lip [41]. Patients can be injected for animation ptosis by using 3–6 units of onabotulinum, incobotulinum, prabotulinum or 9–18 units of abobotulinum. The injection is placed at the base of the anterior nasal spine at the columellar–labial junction in the midline. The results of this injection can be improved with a lateral injection found in the region of the alar portion of the nasalis and the levator labii superioris alaeque nasi muscles. This lateral injection is accomplished with 2–6 units of onabotulinum, incobotulinum, prabotulinum or 6–18 units of concentrated abobotulinum [6, 42]. Improvement of animation ptosis can be seen in this patient who received neurotoxin to the depressor septi nasi, alar portion of the nasalis and the levator labii superioris alaeque nasi (Fig. 11a—frontal view before, b—lateral view before, c—injection technique, d—frontal view after, e—lateral view after).

An additional unique application for neurotoxin in the nose is to address nasal flaring. This occurs in individuals who unconsciously flare their nostrils under stress. This imparts subliminal anger, fear, exhaustion, concern or distress. This is caused by alar nasalis and the medial alar portion of levator labii superioris alaeque nasi [6, 43]. Treatment involves 4–10 units onabotulinum, incobotulinum, prabotulinum or 12–30 units of abobotulinum in the center of each alar rim [6]. Best results are seen in those patients who can actively dilate their nostrils as demonstrated in the before and after photograph of this patient (Fig. 12a, b).

The next area in the midface that can be addressed with neurotoxin is the mouth. Some patients seek consultation for the "canine smile" or "gummy" smile, which is caused by increased tone of the levator labii superioris and levator labii superioris alaeque nasi. These muscles are responsible for elevating the central portion of the lip in relation to the oral commissures, and this results in the gummy smile. The injection point for this muscle is at the apex of the nasolabial fat compartment approximately 2 mm lateral to the

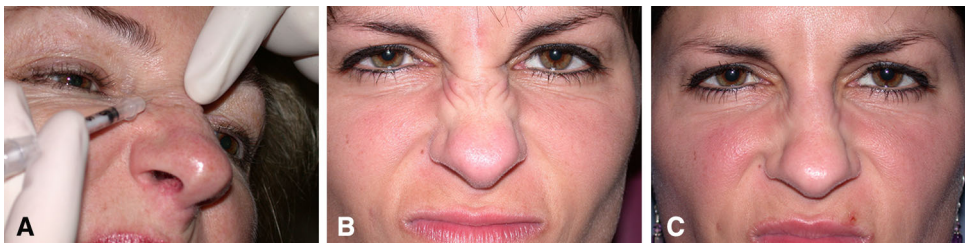


Fig. 10 **a** Injection technique to nasalis muscle to address bunny lines; a patient before (**b**) and after (**c**) treatment with neurotoxin to the nasalis muscle

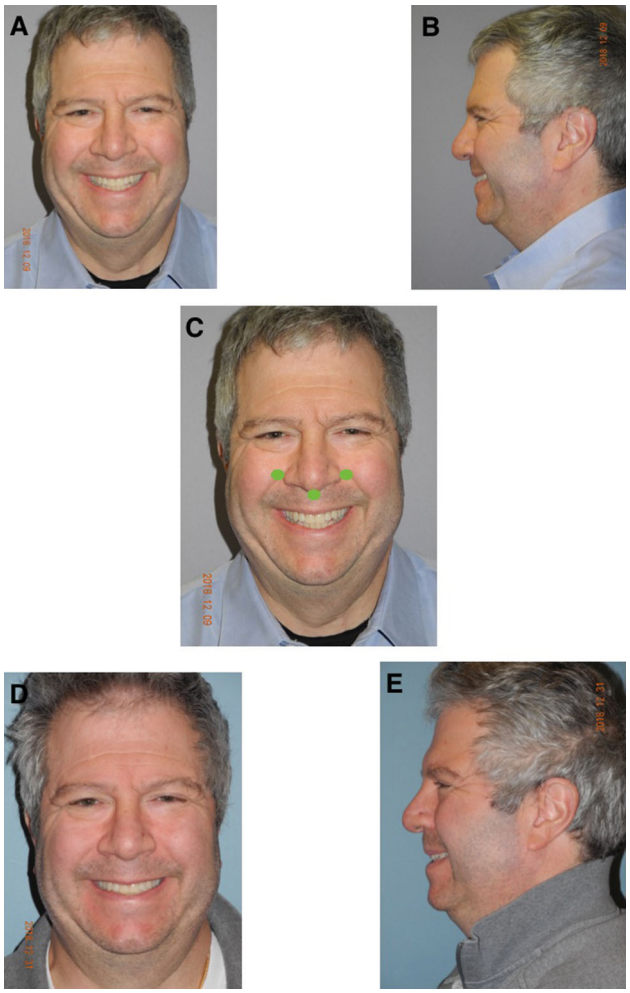


Fig. 11 Treatment of animation ptosis with neurotoxin (**a**—frontal view before, **b**—lateral view before, **c**—injection technique, **d**—frontal view after, **e**—lateral view after)

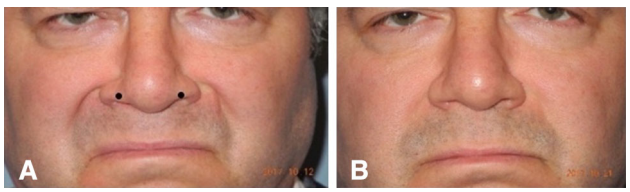


Fig. 12 Before (**a**) and after (**b**) appearance of nasal flaring after treatment with neurotoxin into the center of each alar rim

nasal alar groove. Another injection can be placed 2 mm lateral to this injection point to capture the levator labii superioris [44]. The dosage for the levator labii superioris alaeque nasi is 1–2 units of onabotulinum, incobotulinum, prabotulinum or 3–6 units of concentrated abobotulinum. The dose to the levator labii superioris is 2–4 units of onabotulinum, incobotulinum, prabotulinum or 6–12 units of concentrated abobotulinum to prevent increased spread of effect [6]. The results of this injection can be seen in the before and after pictures of the patient in Fig. 13.

The asymmetric smile can also be treated with a neurotoxin to restore balance [45–46]. As seen in this patient (Fig. 14), the hypertonicity of the levator labii superioris alaeque nasi is not only creating an asymmetry of the upper lip with smiling, but there is also an increase in the depth of the nasolabial fold and elevated posture to the alar lobule accentuated with smiling. In this patient, when the patient smiles before her treatment we can see the three specific issues occurring (Fig. 14a). After one single injection, correction of a unilateral gummy smile, improvement in the depth of the left nasolabial fold and lowering of the left nostril position to provide symmetry are accomplished [6] (Fig. 14b). Smile asymmetry can also be a result of hypertonicity of the depressor anguli oris (DAO). It is responsible for the downward pull of the oral commissure, thus creating several unaesthetic aging characteristics to the lower face such as a sullen or dissatisfied appearance [45–47] (Fig. 15a). This can be successfully addressed with 4 units of onabotulinum, incobotulinum, prabotulinum or 12 units of abobotulinum to each DAO and 4 units to the levator labii superioris alaeque nasi [6, 24] (Fig. 15b). Typically, filler to the nasolabial folds will further help to correct this asymmetry [48]. Caution should be taken because excessive dosing and medial injections may lead to

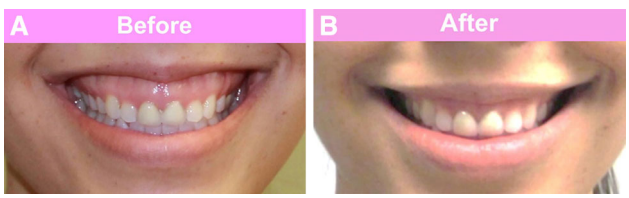


Fig. 13 Treatment of the "gummy smile" with neurotoxin (**a**—before; **b**—after)



Fig. 14 Improvement in upper lip asymmetry, nasolabial fold depth and posture to the nostril (**a**) and after treatment (**b**) of levator labii superioris alaeque nasi hypertonicity with neurotoxin

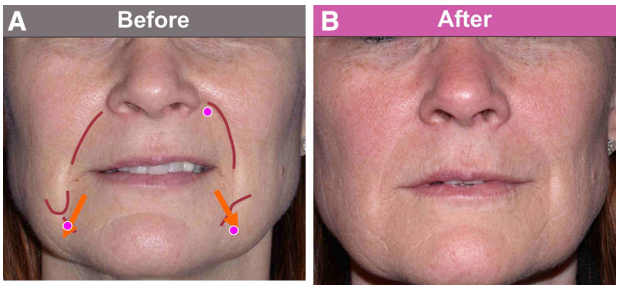


Fig. 15 Treatment of smile asymmetry before (**a**) and after (**b**) treatment with neurotoxin to the depressor anguli oris and levator labii superioris alaeque nasi muscles

paralysis of the depressor labii inferioris muscle, resulting in an asymmetrical smile [47].

Lower Face

The lower face has become a popular and challenging region for neurotoxin therapy. It is an area that benefits primarily from volume replacement and rhytid effacement, with neurotoxins playing an increasing role in its rejuvenation [6]. The central portion of the lower face is dominated by the orbicularis oris, which originates and inserts on itself and is firmly attached to the dermis of the glabrous skin of the lips [49]. Repetitive actions, such as puckering, smoking and drinking through a straw, result in forces being transmitted to the overlying skin subsequently creating rhytids [50]. An effective therapy for treating perioral lines involves the use of 3–4 units of onabotulinum, incobotulinum, prabotulinum or 9–12 units of abobotulinum in the upper lip, and 2 units of onabotulinum, incobotulinum, prabotulinum or 6 units of abobotulinum in the lower lip [6, 24]. It is recommended to deliver one unit aliquots over 4 injection points along the superior vermilion border and 2 injection points along the lower vermilion border, staying 1 cm medial from the oral commissure to avoid unwanted spread into muscles inserting into the modiolus [51]. The positive results of using neurotoxin for perioral rhytids can be seen in the before and after pictures of the patient in Fig. 16a, b.



Fig. 16 Before (**a**) and after (**b**) appearance after treatment of perioral rhytids with neurotoxin

The next area in the lower face frequently injected for aesthetic purposes is the mentalis. This muscle can create dimpling of the skin with elevation (cobblestoning or peau d'orange) as well as a prominent mental crease, which produces an unappealing appearance [23, 47, 52]. Usually, a single injection in the midline of 4–10 units of onabotulinum, incobotulinum, prabotulinum or 12–30 units of abobotulinum can be performed. If one chooses to inject bilaterally on either side of the midline, two injections can be performed with either 4–5 units per injection site with onabotulinum, incobotulinum, prabotulinum or 12–15 units of abobotulinum lateral to the midpoint [6, 24]. These surface changes can be treated with multimodality therapy with fillers and neurotoxins as seen in the before (a) and after (b) photographs of the patient in Fig. 17.

The next muscle to be discussed in the lower one-third of face is the Masseter. This muscle is found at the lateral aspect of the lower face and can create a masculine appearance (hypertrophy) with squaring of the mandible, or feminine appearance with an oval shape to the lower face [53]. Treatment in this region is an example of the powerful effect of facial reshaping with neurotoxins. Hypertrophy may be associated with malocclusion, temporomandibular joint dysfunction or bruxism and may be unilateral or bilateral [54]. This boxy look to the lower one-third of the face creating this masculine appearance can be seen in any ethnic background; however, it is more prevalent in Korean women [55–56].

During treatment, the patient is asked to bite down, creating a firm muscle mass allowing for ease of injection by pressing the hub to the skin. Anywhere between 10–40 units of onabotulinum, incobotulinum, prabotulinum or



Fig. 17 Before (**a**) and after (**b**) appearance after treatment of the mentalis muscle with neurotoxin

30–120 units of abobotulinum are placed [6]. Frequently, patients are seen several weeks later to determine if further injection is necessary. Atrophy can last six to nine months if adequate dosing is applied. Patients who have TMJ symptoms along with masseter hypertrophy often express relief of their symptoms. Injecting in front of the anterior border of the masseter can result in weakness of the risorius with an asymmetric smile and should be avoided [57]. Successful treatment is demonstrated in the before and after picture of this patient (Fig. 18a, b).

Neck

The next body region you can treat with neurotoxin is the neck. The primary muscle of the neck is the platysma, which acts as a major depressor in the lower face [56, 58]. Insertions of this muscle permit it to pull down both the cheek skin and the corners of the mouth [56]. Injecting vertical platysmal bands with neurotoxin negates the downward motion vector of the overlying soft tissue. As a result, a net upward vector is created by the lateral facial extent of the platysma muscle and the midface elevators, thus providing better definition of the jawline [58]. The dosage can be 2.5 units of onabotulinum, incobotulinum, prabotulinum or 7.5 units of abobotulinum every 1.5 cm along the vertical bands of the platysma [6, 59–61] (Fig. 19). Usually multiple bands are found with contraction, and these can be identified and treated. The Nefertiti lift is performed by identifying the important contractile band of the platysma extending into the jowl along with selected injections along the mandible. Another benefit of injecting the medial vertical bands of the platysma muscle is that one can create a sharper cervicomental angle which can be blunted by the presence of these bands [56, 62].

When injecting the platysmal bands, it is best to grasp the contracted band between thumb and index finger and inject tangentially to the surface of the skin, thus preventing the spread of neurotoxin to the deeper neck muscles [6, 59] (Fig. 19). Improper injection technique and/or excessive dosing of neurotoxin (greater than 50 units of onabotulinum, incobotulinum, prabotulinum or 150 units of abobotulinum) in this region may create dysphagia and dysphonia [47, 56, 63]. The successful treatment of platysmal bands with neurotoxin can be seen in the before and after photographs of this patient (Fig. 20a, b).



Fig. 19 Neurotoxin injection technique for platysmal bands

Necklace lines (aka tech lines) are difficult lines to treat with one modality. These lines run horizontally across the neck and are found in flexion creases. These are best treated with neurotoxins and a low viscosity hyaluronic acid. Conservative resurfacing techniques may also be helpful. Injection of 1 unit of onabotulinum, incobotulinum, prabotulinum or 3 units of abobotulinum separated one centimeter along the necklace line starting at the midline of the neck and extending to the medial border of the sternocleidomastoid muscle is complimented by filler injection [64].

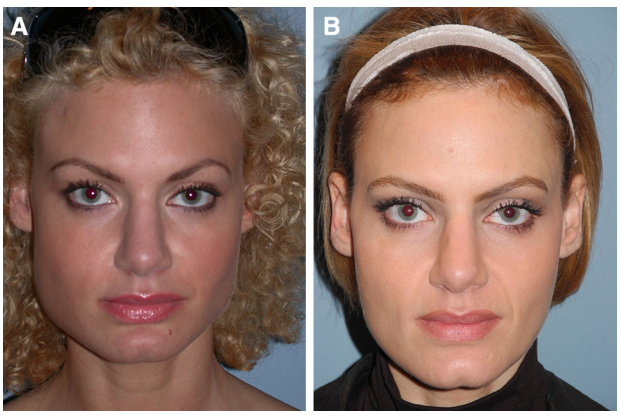


Fig. 18 Before (**a**) and after (**b**) appearance after treatment of the masseter muscle with neurotoxin



Fig. 20 Before (**a**) and after (**b**) appearance after treatment of platysmal bands with neurotoxin

Conclusions

Increasing emphasis has been placed on minimally invasive cosmetic improvement. Treatment goals for botulinum toxin cosmetic applications include recontouring facial proportions, increasing symmetry and maintenance of movement control. Proper patient selection, knowledge of head and neck anatomy, and thorough understanding of injection techniques and properties of each toxin product are needed for optimal outcomes and avoidance of adverse events [6, 65].

Compliance with Ethical Standards

Conflict of interest The authors declare that they have no conflicts of interest to disclose.

Ethical Standards All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was obtained from all participants.

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